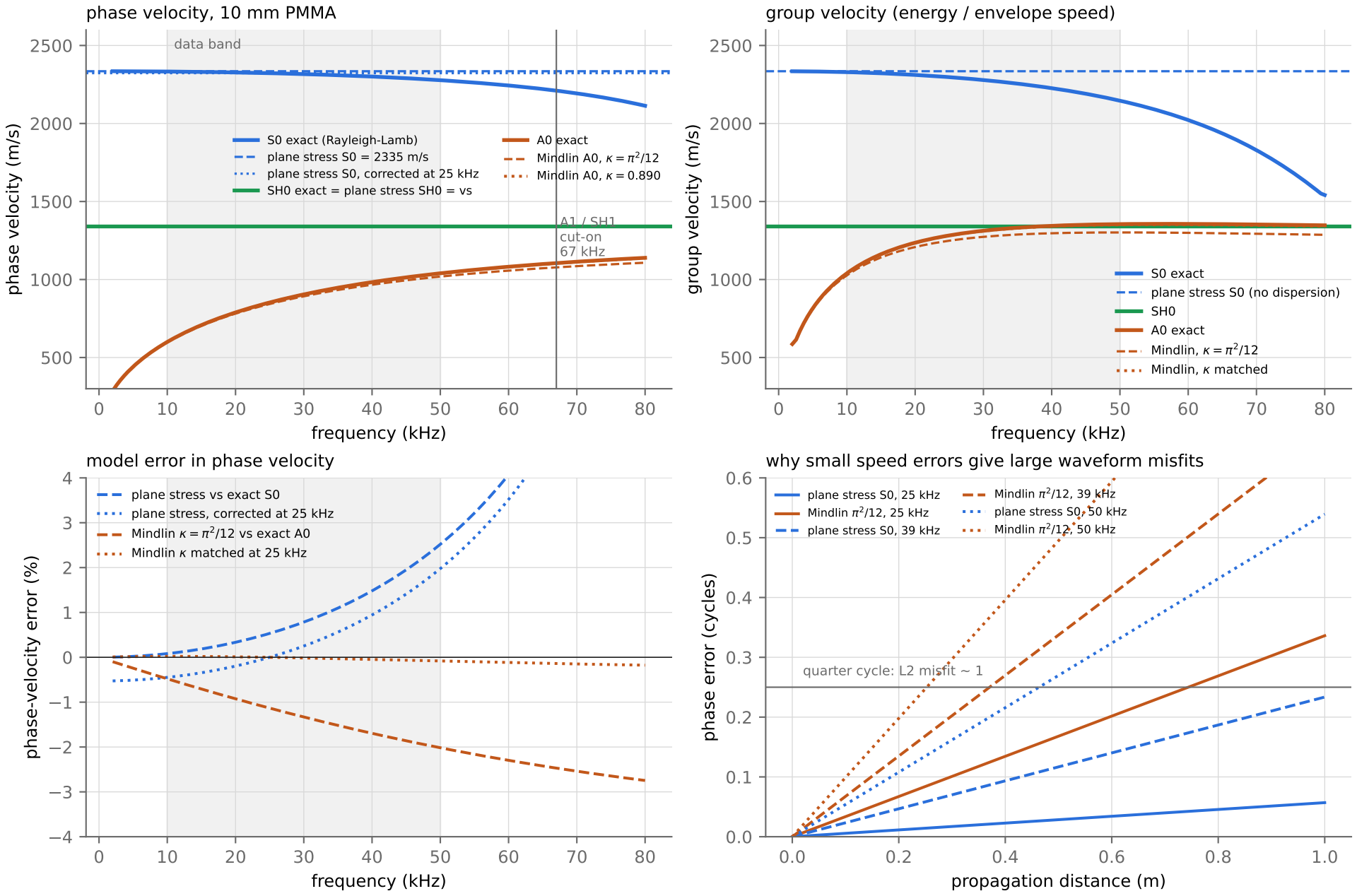
phase velocity, 10 mm PMMA
data band
S0 exact (Rayleigh-Lamb)
plane stress S0 = 2335 m/s
plane stress S0, corrected at 25 kHz
SH0 exact = plane stress SH0 = vs
A0 exact
Mindlin A0, κ = π²/12
Mindlin A0, κ = 0.890
A1 / SH1
cut-on
67 kHz
phase velocity (m/s)
frequency (kHz)
group velocity (energy / envelope speed)
S0 exact
plane stress S0 (no dispersion)
SH0
A0 exact
Mindlin, κ = π²/12
Mindlin, κ matched
group velocity (m/s)
frequency (kHz)
model error in phase velocity
plane stress vs exact S0
plane stress, corrected at 25 kHz
Mindlin κ = π²/12 vs exact A0
Mindlin κ matched at 25 kHz
phase-velocity error (%)
frequency (kHz)
why small speed errors give large waveform misfits
plane stress S0, 25 kHz
Mindlin π²/12, 25 kHz
plane stress S0, 39 kHz
Mindlin π²/12, 39 kHz
plane stress S0, 50 kHz
Mindlin π²/12, 50 kHz
quarter cycle: L2 misfit ~ 1
phase error (cycles)
propagation distance (m)